SENIOR QA ASSESSMENT – MANUAL TESTING

Manual Testing

Feature: Employee Leave Request – requirements review, test design and bug report

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Employee Leave Request – Requirements Review

Feature: Employee Leave Request **Story:** *As an employee, I want to request leave through the HR system, so that I can submit my planned absence for manager approval.*

This document lists the questions, ambiguities, missing requirements, assumptions and risks I would raise with the Product Owner / BA / developers **before** test execution starts.

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1. Questions and ambiguities per acceptance criterion

#	Acceptance criterion	Questions / ambiguities
AC1	Employee can open the Leave Request page	Which roles can access it (ESS user only, Admin, Supervisor on behalf of others)? What is shown if the employee has **no entitlement** for any leave type? Is the page available on mobile?
AC2	Employee can select a Leave Type	Are all leave types listed, or only those with an entitlement / positive balance? Is the **current balance** displayed after selection? Can a leave type be inactive / deleted while a request is open?
AC3	Employee can select From Date and To Date	Are **past dates** allowed (e.g. sick leave reported afterwards)? How far in the future can leave be requested? Can a request span two **leave periods / years**? Is From = To (single day) allowed? Which date format and which calendar first day of week? Is To Date auto-filled from From Date?
AC4	System calculates the number of leave days	Are **weekends** and **public holidays** excluded? Whose work week / holiday calendar is used (employee location, company)? Are **half days / hours / partial days** supported? Is the value recalculated when dates change? Is the calculation done on the client, the server or both (risk of mismatch)? Rounding rules?
AC5	Employee can add an optional comment	Maximum length? Allowed characters / emojis / HTML? Is the comment visible to the manager? Can it be edited after submission?
AC6	Employee can submit the request	What happens when the **balance is insufficient** – block, warn, or allow negative balance? Is the balance **deducted at submission** (pending) or only at approval? Is the manager **notified** (email / in-app)? What if the employee has **no supervisor**? Protection against double submission?
AC7	Submitted request has *Pending Approval* status	Is the status per request or per day (multi-day requests may have mixed statuses after partial approval)? Can a manager auto-approve rules exist (e.g. short leave)?
AC8	Employee can view submitted requests	Which statuses / date range are shown by default? Sorting, filtering, pagination? Which fields (dates, type, days, status, comments, approver)? Are past years visible?
AC9	Employee can cancel a pending request	Only *Pending*, or also *Approved / Scheduled* future leave? Is a confirmation required? Is the balance **restored** immediately? Is the manager notified? Can part of a multi-day request be cancelled?
AC10	System prevents invalid leave requests	"Invalid" is **not defined**. A list of validation rules and messages is needed (see section 2).

2. Missing requirements

1. **Validation rules** (AC10) – at minimum:
   - mandatory fields (leave type, from date, to date);
   - To Date before From Date;
   - request with 0 working days (only weekend / holidays);
   - insufficient balance;
   - **overlapping** with an existing pending / approved request;
   - dates outside the defined leave period;
   - invalid / manually typed dates (e.g. 31 Feb), wrong format;
   - maximum length of the comment.

2. **Balance rules** – when the balance is reduced (pending vs approved), how pending days are shown ("balance" vs "available balance"), restoration on cancel / reject.
3. **Status model** – the business flow only shows *Pending → Approved / Rejected*. Missing states and transitions: **Cancelled**, **Taken** (approved leave in the past), Scheduled, partially approved multi-day requests, cancellation of approved leave, re-submission after rejection.
4. **Manager decision details** – rejection reason mandatory? Can the manager change dates? What happens when the manager is absent (delegation / escalation / SLA)?
5. **Notifications** – who is notified on submit / approve / reject / cancel and through which channel.
6. **Roles and permissions** – an employee must only see and cancel their own requests; can a manager approve their own leave; can an admin apply on behalf of an employee?
7. **Audit trail** – who changed the status and when.
8. **Non-functional** – localisation (date formats, time zones), accessibility, supported browsers / mobile, performance of the list with many requests.

3. Business flow review

Observation	Impact
"Calculate Leave Days" happens before "Valid Request?" but the validation rules that depend on the calculation (0 days, insufficient balance) are not shown.	Unclear where and when the user sees calculation-related errors.
No branch for **Cancel** although AC9 requires it.	State model incomplete; Cancelled status / balance restore untested by design.
No branch for **insufficient balance** or **overlap**.	Most likely production defects live here.
Approved leave has no further state (*Taken*, cancellation of approved leave).	Reporting / balance at year end is unclear.
Manager decision has no timeout / escalation.	Requests may stay pending forever.

4. Assumptions (to be confirmed)

#	Assumption
A1	Work week is Monday–Friday; weekends and configured public holidays are **not** counted.
A2	Full-day leave is the default; partial days (half day / hours) are supported as in OrangeHRM.
A3	Balance shown = entitlement − (approved + pending); pending days are reserved at submission.
A4	Only requests in *Pending Approval* can be cancelled by the employee (as stated in AC9).
A5	Overlapping requests are rejected.
A6	Requests outside the current leave period are not allowed unless the next period is defined.
A7	The manager is the employee's supervisor defined in PIM (*Report-to*).
A8	Dates are stored as calendar dates (no time component), so time zones must not shift them.

5. Risks

#	Risk	Likelihood	Impact	Mitigation / test focus
R1	Day calculation differs between UI and balance deduction (weekends, holidays, inclusive end date).	High	High – financial / payroll impact	Compare displayed days, stored days and balance change for every date pattern.
R2	Time-zone conversion shifts dates (e.g. 12 Aug 00:00 local stored as 11 Aug UTC).	Medium	High	Run with employee / browser / server in different time zones.

#	Risk	Likelihood	Impact	Mitigation / test focus
R3	Overlapping or duplicate requests (double click, two tabs).	Medium	Medium	Concurrency and idempotency tests.
R4	Balance not restored after cancel / reject.	Medium	High	Check balance after each state transition.
R5	Employee can see / cancel another employee's request by changing an id in the URL / API.	Low	High (security, privacy)	Authorisation tests on UI and API.
R6	Leave spanning two leave periods / years deducted from the wrong period.	Medium	Medium	Boundary tests at period end.
R7	Manager not notified → request never approved.	Medium	Medium	Verify notifications / pending list of the supervisor.
R8	Locale-specific date formats cause wrong parsing (dd-mm vs mm-dd).	Medium	Medium	Test with several configured date formats.

6. Entry criteria proposed

- Answers to the open questions above (at least validation rules, balance rules, status model).
- Test environment with: employees with and without supervisors, several leave types, entitlements, a holiday calendar and a defined work week.
- Test users for the roles Employee, Supervisor and Admin.

Employee Leave Request – Test Design

1. Approach

Item	Description
Scope	Apply leave, day calculation, submission, My Leave list, cancellation, validation, manager decision (as far as it affects the employee), balance.
Out of scope	Leave configuration screens (types, periods, holidays) except as test-data set-up; payroll export.
Techniques	Equivalence partitioning and boundary value analysis (dates, balance, comment length), decision table (validation), state transition (request status), role-based testing, error guessing, exploratory testing (charters).
Levels	UI (functional), API checks for authorisation and stored values, cross-browser smoke.
Priorities	**P1** – blocks release / money-related, **P2** – important functionality, **P3** – minor / cosmetic.

Test data

Data	Values
Users	`EMP_A` (has supervisor `MGR_A`), `EMP_B` (no supervisor), `MGR_A`, `ADMIN`
Leave types	Annual (entitlement 10 days), Sick (entitlement 5), Unpaid (entitlement 0), one inactive type
Calendar	Work week Mon–Fri; public holiday on a Wednesday inside the test month
Leave period	1 Jan – 31 Dec
Date patterns	single weekday · Mon–Wed · Thu–Mon (spans weekend) · Sat–Sun only · range containing the holiday · last working day of the period → first of the next

2. Scenario overview (traceability)

Scenario	AC	Test cases
S1 Open Leave Request page	AC1	TC-01 – TC-03
S2 Select leave type and see balance	AC2	TC-04 – TC-07
S3 Select dates	AC3	TC-08 – TC-12
S4 Calculate leave days	AC4	TC-13 – TC-20
S5 Comment	AC5	TC-21 – TC-24
S6 Submit request	AC6, AC7	TC-25 – TC-30
S7 View submitted requests	AC8	TC-31 – TC-34
S8 Cancel pending request	AC9	TC-35 – TC-39
S9 Invalid requests	AC10	TC-40 – TC-49
S10 Manager decision & status flow	Business flow	TC-50 – TC-55
S11 Security / roles	AC1, AC8, AC9	TC-56 – TC-59
S12 Non-functional / compatibility	–	TC-60 – TC-64

3. Test cases

S1 – Open Leave Request page

ID	Title	Preconditions	Steps	Expected result	Priority
TC-01	Employee opens Leave Request page from the menu	`EMP_A` logged in	1. Click Leave → Apply	Page *Apply Leave* is displayed with Leave Type, From Date, To Date, Comment and Apply button	P1
TC-02	Page when employee has no entitlement	`EMP_C` with no entitlements	Open Leave → Apply	Informative message (e.g. "No Leave Types with Leave Balance"), no form, no error	P2
TC-03	Direct URL access when logged out	Logged out	Open the Apply Leave URL	Redirected to login; after login user lands on the requested page or dashboard	P2

S2 – Leave type and balance

ID	Title	Preconditions	Steps	Expected result	Priority
TC-04	Leave type list content	`EMP_A`	Open the Leave Type dropdown	Only active leave types the employee is entitled to are listed; inactive type is not listed	P2
TC-05	Balance displayed for the selected type	Annual balance 10	Select Annual	Balance "10.00 Day(s)" is displayed	P1
TC-06	Balance details popup	–	Click the balance details link	Entitlement, taken, scheduled, pending and balance values add up	P2
TC-07	Changing the type updates the balance	–	Select Annual, then Sick	Balance changes to the Sick balance	P2

S3 – Dates

ID	Title	Preconditions	Steps	Expected result	Priority
TC-08	Select dates with the date picker	–	Pick From and To dates in the calendar	Dates are filled in the configured format	P1
TC-09	Type dates manually	–	Type valid dates in the configured format	Dates are accepted	P2
TC-10	To Date defaults to From Date	–	Select only From Date	To Date is set to the same date; duration "Full Day" is offered	P3
TC-11	Single-day request (From = To)	–	Select the same weekday for both	Accepted; 1 day calculated	P1
TC-12	Range across two leave periods	Period ends 31 Dec	From 30 Dec to 2 Jan	Behaviour as specified (rejected, or split per period); balance deducted from the correct period	P2

S4 – Leave day calculation

ID	Title	Data	Expected result	Priority
TC-13	Consecutive weekdays	Mon–Wed	3.00 days displayed, 3.00 deducted from balance	P1
TC-14	Range spanning a weekend	Thu–Mon	3.00 days (Sat, Sun excluded)	P1
TC-15	Only weekend days	Sat–Sun	0 days → request is blocked with a message	P1
TC-16	Range containing a public holiday	Mon–Fri with holiday on Wed	4.00 days	P1

ID	Title	Data	Expected result	Priority
TC-17	Half day	Single day, Half Day – Morning	0.50 days	P2
TC-18	Partial days on multi-day request	Mon–Wed, *Start Day Only* half day	2.50 days	P2
TC-19	Specific time	Single day, 09:00–13:00	0.50 days (4 h of an 8 h day)	P3
TC-20	Recalculation on date change	Change To Date after calculation	Number of days updates immediately; the value saved equals the value displayed	P1

> For every TC in S4 the check is three-fold: **displayed days = days stored on the request = balance change**.

S5 – Comment

ID	Title	Data	Expected result	Priority
TC-21	Submit without comment	empty	Request submitted (comment is optional)	P1
TC-22	Comment at maximum length	250 characters (or the defined limit)	Accepted and fully displayed	P2
TC-23	Comment over maximum length	limit + 1	Validation message, request not submitted	P2
TC-24	Special characters / script	`<script>alert(1)</script>`, Arabic text, emoji	Stored and displayed as plain text, no script executed	P2

S6 – Submit request

ID	Title	Preconditions	Steps	Expected result	Priority
TC-25	Submit a valid request	Annual balance 10	Select Annual, Mon–Wed, comment, click Apply	Success message; request created with status *Pending Approval (3.00)*	P1
TC-26	Balance after submission	as TC-25	Re-open Apply, select Annual	Available balance reduced by the pending 3 days (or as defined by the balance rules)	P1
TC-27	Supervisor sees the request	`MGR_A`	Log in as `MGR_A`, open Leave List	Request of `EMP_A` is listed as Pending Approval	P1
TC-28	Notification sent	Email configured	Submit	Supervisor (and employee) receive the configured notification	P2
TC-29	Double submission	–	Double-click Apply / press Enter twice	Only one request is created	P2
TC-30	Employee without supervisor	`EMP_B`	Submit a valid request	Behaviour as specified (admin approval, warning); request is not lost	P2

S7 – View submitted requests

ID	Title	Steps	Expected result	Priority
TC-31	Request appears in My Leave	Open Leave → My Leave	Row with correct dates, type, days, status, comment	P1
TC-32	Filter by date range and status	Filter Pending only, then Cancelled only	Only matching rows; record count matches	P2
TC-33	Multi-day request details	Open the details of a Mon–Wed request	One line per day with individual status	P3
TC-34	Many requests	Employee with > 50 requests	Pagination / sorting work, no missing rows	P3

S8 – Cancel a pending request

ID	Title	Steps	Expected result	Priority
TC-35	Cancel a pending request	My Leave → Cancel on a pending row	Success message; status *Cancelled*	P1
TC-36	Balance restored after cancel	Check balance after TC-35	Balance equals the value before the request	P1
TC-37	Cancel not offered for rejected request	Request rejected by manager	No Cancel action for the employee	P2
TC-38	Cancel approved future leave	Request approved	Behaviour as specified (allowed / blocked / needs approval)	P2
TC-39	Manager acts while employee cancels	Two sessions: manager approves, employee cancels at the same time	Final state is consistent; balance correct; no error page	P2

S9 – Invalid requests (decision table)

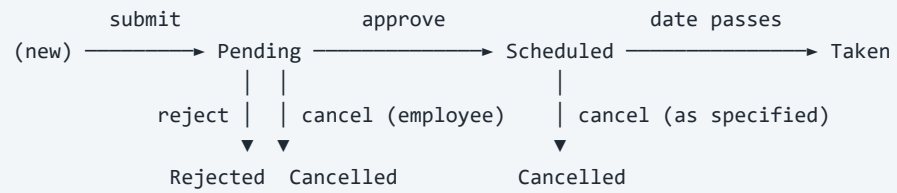
Rule	Leave type	Dates	Balance	Overlap	Expected
R1	missing	valid	ok	no	"Required" on Leave Type
R2	ok	From missing	ok	no	"Required" on From Date
R3	ok	To < From	ok	no	"To date should be after from date"
R4	ok	only weekend / holidays	ok	no	Blocked – 0 working days
R5	ok	valid	insufficient	no	Blocked or warning as specified
R6	ok	valid	ok	yes	Blocked – overlapping request shown
R7	ok	invalid date (31 Feb, wrong format)	ok	no	"Should be a valid date in yyyy-dd-mm format"
R8	ok	outside leave period	ok	no	Blocked – leave period not defined

ID	Title	Rule	Priority
TC-40	Submit without leave type	R1	P1
TC-41	Submit without From Date	R2	P1
TC-42	To Date before From Date	R3	P1
TC-43	Request containing only weekend days	R4	P1
TC-44	Request exceeding balance (balance 2, request 3 days)	R5	P1
TC-45	Request exactly equal to balance (boundary)	R5 boundary	P1
TC-46	Request overlapping an existing pending request	R6	P1
TC-47	Request overlapping an approved request (one day overlap)	R6	P2
TC-48	Invalid / mistyped date	R7	P2
TC-49	Date outside the leave period	R8	P2

For each invalid case also verify: no success message, nothing new in My Leave, balance unchanged.

S10 – Manager decision and status flow

State transitions to cover:



ID	Title	Expected result	Priority
TC-50	Manager approves	Status *Scheduled* (approved); employee sees the new status; balance final	P1
TC-51	Manager rejects	Status *Rejected*; balance restored; employee sees the rejection (and reason if required)	P1
TC-52	Partial approval of a multi-day request	Per-day statuses and balance are consistent	P2
TC-53	Approved leave in the past	Status changes to *Taken*	P3
TC-54	Invalid transition	Approving an already cancelled request is not possible (UI and API)	P2
TC-55	Manager cannot approve own request	Own request does not appear as actionable for self-approval	P2

S11 – Security and roles

ID	Title	Expected result	Priority
TC-56	Employee sees only own requests	Another employee's requests are not listed	P1
TC-57	Cancel another employee's request through the API / URL	Request rejected with 401/403; data unchanged	P1
TC-58	Employee cannot approve own request through the API	Request rejected	P1
TC-59	Session timeout during submission	User is asked to log in again; no partial request created	P3

S12 – Non-functional / compatibility

ID	Title	Expected result	Priority
TC-60	Different date formats configured (dd-mm-yyyy, mm-dd-yyyy, yyyy-dd-mm)	Dates parsed and displayed correctly; no day / month swap	P1
TC-61	Browser / server in different time zones (e.g. UTC-8 browser, UTC+3 server)	Stored dates equal selected dates; no off-by-one	P1
TC-62	Cross-browser smoke (Chrome, Firefox, Edge, Safari)	Apply / view / cancel work	P2
TC-63	Mobile viewport	Form usable, date picker usable	P3
TC-64	Keyboard-only and screen reader	Fields have labels, errors are announced	P3

4. Exploratory testing charters

Charter	Focus
EX-1	Explore the day calculation with unusual ranges (holiday on a weekend, leap day 29 Feb, end of year) to discover mismatches between displayed and deducted days.
EX-2	Explore concurrent actions (two tabs, back button after submit, refresh during submit) to discover duplicate or lost requests.
EX-3	Explore balance behaviour across the whole life cycle (submit → reject → resubmit → approve → cancel) to discover balances that are not restored.
EX-4	Explore configuration changes while requests are pending (holiday added, work week changed, entitlement reduced) to discover stale calculations.

5. Automation candidates

Regression-worthy and stable: TC-11, TC-13, TC-14, TC-16, TC-25, TC-26, TC-31, TC-35, TC-36, TC-40 – TC-46, TC-50, TC-51, TC-56 – TC-58 (the last three at API level). Several of these are already automated in this repository (`LeaveTest`, `NegativeValidationTest`).

Bug Report

Field	Value
ID	LEAVE-001
Title	Leave balance is reduced by 4 days for a 3-day leave request (10–12 Aug) – displayed duration and deducted days do not match
Component	Leave → Apply Leave / Leave balance
Environment	QA – OrangeHRM OS 5.9 (build / commit: *to be filled*), Chrome 129 / Windows 11
Reported by / date	QA – *date of finding*
Severity	**Major** – incorrect leave balance (affects employee entitlements and potentially payroll)
Priority	**High** – must be fixed before release
Reproducibility	*To be confirmed* (e.g. 3/3 attempts with the data below)
Status	New

Preconditions

- Employee `EMP_A` logged in as ESS user, supervisor `MGR_A`.
- Leave type **Annual Leave**, entitlement 10.00 days for the current leave period, no other requests → balance **10.00**.
- Work week Monday–Friday; no public holiday between 10 and 12 August.
- 10 Aug 2026 = Monday, 12 Aug 2026 = Wednesday (if the year is different, record the weekdays).

Steps to reproduce

1. Log in as `EMP_A`.
2. Go to **Leave** → **Apply**.
3. Select Leave Type **Annual Leave** – note the displayed balance (**10.00 Day(s)**).
4. Set **From Date = 10 Aug 2026**, **To Date = 12 Aug 2026**, Partial Days = None (full days).
5. Note the calculated duration shown on the form.
6. Click **Apply**.
7. Go to **Leave** → **My Leave** and open the request.
8. Go back to **Leave** → **Apply**, select **Annual Leave** and check the balance (or open *Leave → Entitlements → My Leave Entitlements and Usage Report*).

Expected result

- Step 5 / 7: the request shows **3.00** days (Mon, Tue, Wed).
- Step 8: the balance is **7.00** days (10.00 − 3.00).
- Displayed days, days stored on the request and days deducted are identical.

Actual result

- Step 5 / 7: the request shows **3.00** days.
- Step 8: the balance is **6.00** days – **4.00** days were deducted.

Evidence (attach)

- Screenshot of the Apply form with the 3-day calculation.
- Screenshot of My Leave (request row and the per-day details).
- Screenshot of the balance before and after the request.
- Network capture (HAR) of the *apply leave* request and of the balance / entitlement responses.
- If available: database rows for the request (`ohrm_leave_request`, `ohrm_leave` – number of day rows and their `length_days`, `ohrm_leave_entitlement.days_used`).
- Browser time zone and server time zone.

Impact

- Employees lose leave days they did not take; the error accumulates with every request.
- The balance shown to the employee and the request duration contradict each other, which damages trust and generates HR tickets.
- If the balance feeds payroll or leave encashment, it has a financial impact.

Notes for investigation

Observations that would help narrow the root cause (to be checked, not confirmed):

Hypothesis	How to check
End date counted twice / off-by-one – deduction counts `to - from + 1` on top of an inclusive range, or iterates `from..to` inclusive while the UI uses working days.	Repeat with a single day (10 Aug → 10 Aug). Expected 1, check if 2 is deducted.
Time-zone shift – dates are stored as date-time and converted to UTC, so 12 Aug becomes 13 Aug or 10 Aug becomes 9 Aug in the deduction logic.	Repeat with browser / user / server set to the same time zone, then different ones (e.g. UTC+3 vs UTC). Check stored dates in the request details and API payload.
Different calendar in the two calculations – the UI excludes a non-working day / holiday but the deduction does not (or the reverse).	Check work week and holiday configuration for the employee's location; repeat with a range Thu–Mon.
Partial-day setting ignored in the deduction (e.g. "All days – half day" treated as full + extra).	Repeat with Partial Days = None vs Half Day.
Double deduction – pending days reserved at submission and deducted again at approval / by a scheduled job.	Compare balance right after submission, after approval, and after the next scheduled job run.
Stale balance cache	Re-login / clear cache and check the balance again; compare with the entitlement usage report.

Additional data to collect: whether other leave types and other employees are affected, whether the extra day appears in the request's per-day list (4 rows instead of 3), and the first build where it occurs (regression vs existing issue).

Suggested regression tests after the fix

- Single day, Mon–Wed, range over a weekend, range over a holiday, half day, partial days.
- Balance check after submit, approve, reject and cancel.
- Same scenarios with browser and server in different time zones.